Numerical Integration Using Polynomial Regression

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Abstract

Numerical integration is an important tool for solving integrals when an anti-derivative cannot be found. Approximation techniques, such as the Trapezoidal rule, Simpson's rule, or Boole's rule find solutions to integrals through computation. In this paper, we outline a family of techniques for numerical integration using polynomial regressions of sampled points from the function and derive the error bound for the linear case.

1 Introduction

Let $f(x)$ be a function whose definite integral we want to approximate numerically. To approximate the integral, we will partition $f(x)$ into intervals, and will use the ordered pairs at the endpoints of these partitions to derive a set of polynomial regressions over the intervals which can be easily integrated to get a numerical approximation of $f(x)$. This system defines a family of approximations, one per degree of regression *i.e linear, quadratic, cubic, etc.* which can be used to get arbitrary precision.

One motivation for taking this approach is that this approximation is guaranteed to have the least-squares property, meaning the underlying regression that we are integrating to get our numerical approximation is the *best-fitting* polynomial of its degree of the sampled points. In this paper, we will derive the linear case while outlining the steps that can be used to derive the formulas for higher-degree polynomial regressions.

2 Projecting Three Points Onto LSRL

Let x_i, x_{i+1}, x_{i+2} be the three points that define two equally-sized intervals over which we want to approximate $\int f(x)dx$. Consider a hypothetical linear regression of these points and their function values, $\hat{f}(x)$, such that each $f(x_i)$ is exactly described by $\hat{f}(x_i)$. In such a regression, each point inputted into the Slope-Intercept formula for a line, $y = mx + b$, with m and b being constants

derived from our regression of the points, will be exactly equal to the function value at said point.

$$\begin{aligned} f(x_i) &= mx_i + b \\ f(x_{i+1}) &= mx_{i+1} + b \\ f(x_{i+2}) &= mx_{i+2} + b \end{aligned}$$

We can understand this system of equations as denoting the desired equality between $\hat{f}(x)$ and $f(x)$. By itself, such a system of equations has no solution, unless the three points happen to fall on a line. However, we can *normalize* the equations, resulting in another system of equations that describe a true linear regression of the points $\hat{f}(x)$ which is, by definition, the closest projection of the set of points onto a LSRL.

We can express the desired relationship between the function and the LSRL as a linear system in the form $\vec{\mathbf{v}} = A\vec{\mathbf{x}}$. To project the points onto the LSRL, we can normalize this linear system by multiplying both sides by the transposed matrix A^T, giving:

$$\begin{bmatrix} x_i & x_{i+1} & x_{i+2} \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} f(x_i) \\ f(x_{i+1}) \\ f(x_{i+2}) \end{bmatrix} = \begin{bmatrix} x_i & x_{i+1} & x_{i+2} \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_i & 1 \\ x_{i+1} & 1 \\ x_{i+2} & 1 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix}$$

$$\begin{bmatrix} x_i f(x_i) + x_{i+1} f(x_{i+1}) + x_{i+2} f(x_{i+2}) \\ f(x_i) + f(x_{i+1}) + f(x_{i+2}) \end{bmatrix} = \begin{bmatrix} x_i^2 + x_{i+1}^2 + x_{i+2}^2 & x_i + x_{i+1} + x_{i+2} \\ x_i + x_{i+1} + x_{i+2} & 3 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix}$$

To simplify our linear system, note that the sum of three evenly spaced numbers gives 3 times the middle number. Applying this concept to our partition of the function gives $x_i + x_{i+1} + x_{i+2} = 3x_{i+1}$. Additionally, squaring this equality and solving for $x_i^2 + x_{i+1}^2 + x_{i+2}^2$ gives:

$$\begin{aligned} (x_i + x_{i+1} + x_{i+2})^2 &= x_i^2 + x_{i+1}^2 + x_{i+2}^2 + 2(x_i x_{i+1} + x_{i+1} x_{i+2} + x_i x_{i+2}) \\ x_i^2 + x_{i+1}^2 + x_{i+2}^2 &= (x_i + x_{i+1} + x_{i+2})^2 - 2(x_i x_{i+1} + x_{i+1} x_{i+2} + x_i x_{i+2}) \\ x_i^2 + x_{i+1}^2 + x_{i+2}^2 &= (x_i + x_{i+1} + x_{i+2})^2 - 2x_{i+1}(x_i + x_{i+2}) - 2x_i x_{i+2} \\ x_i^2 + x_{i+1}^2 + x_{i+2}^2 &= 9x_{i+1}^2 - 4x_{i+1}^2 - 2x_i x_{i+2} \\ x_i^2 + x_{i+1}^2 + x_{i+2}^2 &= 5x_{i+1}^2 - 2x_i x_{i+2} \end{aligned}$$

Unfortunately, there is no simplification that can be done for the elements of the vector on the left hand side of our linear system. However, plugging in the expressions $3x_{i+1}$ and $5x_{i+1}^2 - 2x_i x_{i+2}$ back into the original matrix gives:

$$\begin{bmatrix} x_i f(x_i) + x_{i+1} f(x_{i+1}) + x_{i+2} f(x_{i+2}) \\ f(x_i) + f(x_{i+1}) + f(x_{i+2}) \end{bmatrix} = \begin{bmatrix} 5x_{i+1}^2 - 2x_i x_{i+2} & 3x_{i+1} \\ 3x_{i+1} & 3 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix}$$

By normalizing the linear system, we guaranteed that the resultant linear system will be solvable. Solving will give expressions for m and b, giving the

full equation for $\hat{f}$, which can be integrated to find the approximate integral for every two sections. We can solve for m and b using Cramer's rule.

Starting with finding m, Cramer's rule outlines that in order to find m's expression, we take the matrix A and replace first column with the vector $\vec{\mathbf{v}}$ on the left hand side of the equation. We must then take the determinant of this new matrix and divide it by the determinant of the original matrix A.

$$m = \frac{\begin{vmatrix} x_i f(x_i) + x_{i+1} f(x_{i+1}) + x_{i+2} f(x_{i+2}) & 3x_{i+1} \\ f(x_i) + f(x_{i+1}) + f(x_{i+2}) & 3 \end{vmatrix}}{\begin{vmatrix} 5x_{i+1}^2 - 2x_i x_{i+2} & 3x_{i+1} \\ 3x_{i+1} & 3 \end{vmatrix}}$$

By expanding out the determinant formulas and simplifying the expression, we can get an expression for m. Note that during simplification, $x_{i+2} - x_{i+1} = x_{i+1} - x_i = \Delta x$.

$$m = \frac{3(x_i f(x_i) + x_{i+1} f(x_{i+1}) + x_{i+2} f(x_{i+2})) - 3x_{i+1}(f(x_i) + f(x_{i+1}) + f(x_{i+2}))}{3(5x_{i+1}^2 - 2x_i x_{i+2}) - (3x_{i+1})^2}$$

$$m = \frac{3(x_i f(x_i) + x_{i+2} f(x_{i+2})) - 3x_{i+1}(f(x_i) + f(x_{i+2}))}{15x_{i+1}^2 - 6x_i x_{i+2} - 9x_{i+1}^2}$$

$$m = \frac{3x_i f(x_i) + 3x_{i+2} f(x_{i+2}) - 3x_{i+1} f(x_i) - 3x_{i+1} f(x_{i+2})}{6x_{i+1}^2 - 6x_i x_{i+2}}$$

$$m = \frac{x_i f(x_i) + x_{i+2} f(x_{i+2}) - x_{i+1} f(x_i) - x_{i+1} f(x_{i+2})}{2x_{i+1}^2 - 2x_i x_{i+2}}$$

$$m = \frac{f(x_i)(x_i - x_{i+1}) + f(x_{i+2})(x_{i+2} - x_{i+1})}{2x_{i+1}^2 - 2x_i x_{i+2}}$$

$$m = \frac{f(x_{i+2})(x_{i+2} - x_{i+1}) - f(x_i)(x_{i+1} - x_i)}{2x_{i+1}^2 - 2x_i x_{i+2}}$$

$$m = \frac{\Delta x}{2} \frac{f(x_{i+2}) - f(x_i)}{x_{i+1}^2 - x_i x_{i+2}}$$

Similarly to finding m, we can solve for b using Cramer's rule as well. To do so, we take the matrix A and replace the second column with the vector $\vec{\mathbf{v}}$ on the left hand side of the equation. We must then take the determinant of this new matrix and divide it by the determinant of the original matrix A.

$$b = \frac{\begin{vmatrix} 5x_{i+1}^2 - 2x_i x_{i+1} & x_i f(x_i) + x_{i+1} f(x_{i+1}) + x_{i+2} f(x_{i+2}) \\ 3x_{i+1} & f(x_i) + f(x_{i+1}) + f(x_{i+2}) \end{vmatrix}}{\begin{vmatrix} 5x_{i+1}^2 - 2x_i x_{i+2} & 3x_{i+1} \\ 3x_{i+1} & 3 \end{vmatrix}}$$

Exactly like in solving for m, by expanding out the determinant formulas and simplifying the expression, we can get an expression for b. Note that during simplification, $x_{i+2} - x_{i+1} = x_{i+1} - x_i = \Delta x$.

$$b = \frac{(5x_{i+1}^2 - 2x_i x_{i+2})(f(x_i) + f(x_{i+1}) + f(x_{i+2})}{3(5x_{i+1}^2 - 2x_i x_{i+2}) - (3x_{i+1})^2} - \frac{3x_{i+1}(x_i f(x_i) + x_{i+1} f(x_{i+1}) + x_{i+2} f(x_{i+2}))}{3(5x_{i+1}^2 - 2x_i x_{i+2}) - (3x_{i+1})^2}$$

$$b = \frac{(5x_{i+1}^2 - 2x_i x_{i+2})(f(x_i) + f(x_{i+2}))}{6x_{i+1}^2 - 6x_i x_{i+2} 2} - \frac{3x_{i+1}(x_i f(x_i) + x_{i+2} f(x_{i+2}))}{6x_{i+1}^2 - 6x_i x_{i+2}} + \frac{(5x_{i+1}^2 - 3x_{i+1}^2 - 2x_i x_{i+1}) f(x_{i+1})}{6x_{i+1}^2 - 6x_i x_{i+2}}$$

$$b = \frac{(3x_{x+1}^2 + 2x_{i+1}^2 - 2x_i x_{i+2})(f(x_i) + f(x_{i+2}))}{6x_{i+1}^2 - 6x_i x_{i+2}} - \frac{3x_{i+1}(x_i f(x_i) + x_{i+2} f(x_{i+2}))}{6x_{i+1}^2 - 6x_i x_{i+2}} + \frac{(2x_{i+1}^2 - 2x_i x_{i+1}) f(x_{i+1})}{6x_{i+1}^2 - 6x_i x_{i+2}}$$

$$b = \frac{3x_{i+1}^2 (f(x_i) + f(x_{i+2})) - 3x_{i+1}(x_i f(x_i) + x_{i+2} f(x_{i+2}))}{6x_{i+1}^2 - 6x_i x_{i+2}} + \frac{(2x_{i+1}^2 - 2x_i x_{i+2})(f(x_i) + f(x_{i+2}))}{6x_{i+1}^2 - 6x_i x_{i+2}} + \frac{f(x_{i+1})}{3}$$

$$b = \frac{3x_{i+1}^2 f(x_i) + 3x_{i+1}^2 f(x_{i+2}) - 3x_i x_{i+1} f(x_i) - 3x_{i+1} x_{i+2} f(x_{i+2})}{6x_{i+1}^2 - 6x_i x_{i+2}} + \frac{(f(x_i) + f(x_{i+1}) + f(x_{i+2}))}{3}$$

$$b = \frac{(x_{i+1}^2 - x_i x_{i+1}) f(x_i) + (x_{i+1}^2 - x_{i+1} x_{i+2}) f(x_{i+2})}{2x_{i+1}^2 - 2x_i x_{i+2}} + \frac{(f(x_i) + f(x_{i+1}) + f(x_{i+2}))}{3}$$

$$b = x_{i+1} \frac{(x_{i+1} - x_i) f(x_i) + (x_{i+1} - x_{i+2}) f(x_{i+2})}{2x_{i+1}^2 - 2x_i x_{i+2}} + \frac{(f(x_i) + f(x_{i+1}) + f(x_{i+2}))}{3}$$

Note that in the first term in our expression for b at this point, we can factor out a $\frac{\Delta x}{2}$ from the expression which leaves, making it identical to the term for m, which we can replace to get the final expression for b.

$$b = x_{i+1}\frac{\Delta x(f(x_i) - f(x_{i+2}))}{2x_{i+1}^2 - 2x_i x_{i+2}} + \frac{(f(x_i) + f(x_{i+1}) + f(x_{i+2}))}{3}$$

$$b = -x_{i+1}\frac{\Delta x}{2}\frac{f(x_{i+2}) - f(x_i)}{x_{i+1}^2 - x_i x_{i+2}} + \frac{(f(x_i) + f(x_{i+1}) + f(x_{i+2}))}{3}$$

$$b = \frac{f(x_i) + f(x_{i+1}) + f(x_{i+2})}{3} - x_{i+1}m$$

At this point, we have found the formula for $\hat{f}(x)$, which is by definition the best possible fit for the sampled points. Because $\hat{f}(x)$ is a linear function, it is easy to integrate over the sampled interval, giving us an approximation of the integral of the original function $f(x)$. Thus, we integrate $\hat{f}(x)$ over the interval from x_i to x_{i+1}.

$$\int_{x_i}^{x_{i+2}} (mx + b)dx$$

$$\frac{m}{2}(x_{i+2}^2 - x_i^2) + b(x_{i+2} - x_i)$$

$$\frac{m}{2}(x_{i+2} - x_i)(x_{i+2} + x_i) + b(x_{i+2} - x_i)$$

Recall that $x_i + x_{i+1} + x_{i+2} = 3x_{i+1}$ and that, by extension, $x_i + x_{i+2} = 2x_{i+1}$. This, along with b and Δx can be substituted into our expression to get a new expression in terms of the original inputs and their respective function values. Note that m is not substituted in because it is present in the expression for b, which will lead to a cancellation.

$$4x_{i+1}\Delta x\frac{m}{2} + 2\Delta x(\frac{f(x_i) + f(x_{i+1}) + f(x_{i+2})}{3} + x_{i+1}m)$$

$$2x_{i+1}\Delta xm + \frac{2\Delta x}{3}(f(x_i) + f(x_{i+1}) + f(x_{i+2})) + 2x_{i+1}\Delta xm$$

After canceling, we are left with a formula for the integral of our linear regression $\hat{f}(x)$, which we will call N, which approximates the integral of the original function $f(x)$. Note that N only gives the approximation over two of the total number of intervals, using only three of the points.

$$N = \frac{2\Delta x}{3}(f(x_i) + f(x_{i+1}) + f(x_{i+2}))$$

While this formula gives the approximation for a set of two intervals, to get the full approximation, we must expand this sum to include all the sampled points. This is done by repeating this section every two intervals until all the intervals have been included. Note that Δx can also be replaced with the equivalent $\frac{x_n - x_0}{n}$.

$$N_n = \frac{2(x_n - x_0)}{3n}(f(x_0) + f(x_1) + 2f(x_2) + f(x_3) + 2f(x_4) + ... + f(x_{n-1}) + f(x_n))$$

3 Finding The Error Bound of The Approximation

A natural question arises once the approximation formula - *how accurate is the approximation?* To determine this, we must determine a bound on the error of N_n, which will output, given the interval, function, and number of sub-intervals n, the maximum of $\left|\int f(x)dx - N_n\right|$.

For N_n, note that the error of our approximation of $f(x)$ is dependent on at least the second derivative of the function $f''(x)$. This can be understood by realizing that there can exist a set of three points sampled from $f(x)$ that lie perfectly on a line. In such a case, the approximation would only be without error if and only if $f(x)$ is a linear function. In any other case, however, the curvature of $f(x)$, would result in a difference in the values of $f(x)$ and $\hat{f}(x)$ which would be unaccounted for without the second derivative considered. Practically, this means that we will cancel all terms dependent on $f'(x)$.

To find the error bound for our approximation N_n, we will follow a similar path to the derivation of the error bound for the Trapezoidal rule. Let us start with the integral of $f(x)$. Note that we only need to take the integral of $f(x)$ over two of the sub-intervals because every two sub-intervals is a building block of the entire approximation. This error can be summed over half the number of sub-intervals to account for the full approximation. We want to find a way to pull the formula for N_n out of the integral and cancel out as many terms as possible. This will leave the integral of $f(x)$ in terms of N_n and another integral expression, which will give the error bound formula when evaluated.

Note that N_n is dependent on the function values at all three input values. However, evaluation of the integral of the function is only dependent on the endpoints of the integration. Therefore, we can split the interval into one from x_i to x_{i+1} and from x_{i+1} to x_{i+2}. We can also take a t-substitution, $t = x - x_i$, to get the integral expression in terms of Δx, simplifying evaluation.

$$\int_{x_i}^{x_{i+2}} f(x)dx = \int_0^{\Delta x} f(t + x_i)dt + \int_{\Delta x}^{2\Delta x} f(t + x_i)dt$$

At this point, we can integrate both integrals by parts to create a set of expressions in terms of the function values at the inputs which cancel out, leaving a term for our approximation N_n. For each integral, we can differentiate $f(t + x_i)$ and integrate dt, giving $du = f'(t + x_i)$ and $v = t + C$. Note that because the derivative of a constant is zero, it is legitimate to add constants the expression for v. We will used these constants to algebraically manipulate the resulting expressions to pull out the formula for N_n from the integral.

$$\int_{x_i}^{x_{i+2}} f(x)dx = (t+C_0)f(t+x_i)\Big|_0^{\Delta x} + (t+C_1)f(t+x_i)\Big|_{\Delta x}^{2\Delta x}$$
$$+ \int_0^{\Delta x} (t+C_0)f'(t+x_i)dt + \int_{\Delta x}^{2\Delta x} (t+C_1)f'(t+x_i)dt$$

Let us evaluate the terms outside of integrals. By plugging in the endpoints, we will be left with coefficients by each $f(x)$ term. Note that we want the coefficients to all be equal, which can be ensured by controlling the constants C_0 and C_1.

$$(\Delta x+C_0)f(\Delta x+x_i)-C_0f(x_i)+(2\Delta x+C_1)f(2\Delta x+x_i)-(\Delta x+C_1)f(\Delta x+x_i)$$
$$-C_0f(x_i)+(C_0-C_1)f(x_{i+1})+(2\Delta x+C_1)f(x_{i+2})$$

To solve, we can use the desired equality of all the coefficients to derive the values of each of the constants C_0 and C_1. To do this, we must first set the first two coefficients equal to each other and solve for C_0 in terms of C_1.

$$-C_0 = C_0 - C_1$$
$$C_0 = \frac{1}{2}C_1$$

Now that we have our formula for C_0 in terms of C_1, we can set the first and last coefficients to be equal and substitute our expression in terms of C_1 for C_0. Once we have this equation, we can algebraically solve for C_1's value, which can then be used to find the value of C_0.

$$-\frac{1}{2}C_1 = 2\Delta x + C_1$$
$$-2\Delta x = \frac{3}{2}C_1$$
$$C_1 = -\frac{4}{3}\Delta x$$

Therefore, the two constants C_0 and C_1 which, when plugged into the terms given by integration by parts, gives our approximation N_n over the interval are:

$$C_0 = -\tfrac{2}{3}\Delta x$$
$$C_1 = -\tfrac{4}{3}\Delta x$$

Plugging these constants back into the evaluated terms outputted from the process of integration by parts simplifies into our approximation N_n.

$$\frac{2}{3}\Delta x f(x_i) + (-\frac{2}{3}\Delta x + \frac{4}{3})f(x_{i+1}) + (2\Delta x - \frac{4}{3}\Delta x)f(x_{i+2})$$

$$\frac{2}{3}\Delta x f(x_i) + \frac{2}{3}\Delta x f(x_{i+1}) + \frac{2}{3}\Delta x f(x_{i+2})$$
$$\frac{2\Delta x}{3}(f(x_i) + f(x_{i+1}) + f(x_{i+2}))$$

Returning to the original integral for the error bound formula from $f(x)$, we can replace the terms outside of the integral with $2N_n$.

$$\int_{x_i}^{x_{i+2}} f(x)dx = N_n + \int_0^{\Delta x} (t + C_0)f'(t + x_i)dt + \int_{\Delta x}^{2\Delta x} (t + C_1)f'(t + x_i)dt$$

We can integrate by parts again to produce terms based on $f'(x)$. This is desirable because we want to cancel out as many terms as possible. Similarly to the first iteration of integration by parts, we can set $u = f'(t + x_i)$ and $dv = t + C_0$, giving $du = f''(t + x_i)$ and $v = \frac{1}{2}t^2 + C_0 t + C$. To simplify, we can complete the square on v and substitute $C_2 = -\frac{C_0^2}{2} + C$, giving $v = \frac{(t+C_0)^2}{2} + C_2$ Note that although we only show the case for the first integral with C_0, it must be done separately for both integrals using C_1 for the second integral.

$$\Big(\frac{(t + C_0)^2}{2} + C_2\Big)f'(t + x_i)\Big|_0^{\Delta x}$$
$$+ \Big(\frac{(t + C_1)^2}{2} + C_3\Big)f'(t + x_i)\Big|_{\Delta x}^{2\Delta x} + \int_0^{\Delta x} \Big(\frac{(t + C_0)^2}{2} + C_2\Big)f''(t + x_i)dt$$
$$+ \int_{\Delta x}^{2\Delta x} \Big(\frac{(t + C_1)^2}{2} + C_3\Big)f''(t + x_i)dt$$

Let us again examine the terms outside of the integrals. Our goal with these terms is to choose constants C_2 and C_3 such that all the terms cancel out. To start, we can evaluate the terms and plug in the known constants $C_0 = -\frac{2}{3}\Delta x$ and $C_1 = -\frac{4}{3}\Delta x$.

$$\Big(\frac{(\Delta x + C_0)^2}{2} + C_2\Big)f'(\Delta x + x_i) - \Big(\frac{C_0^2}{2} + C_2\Big)f'(x_i)$$
$$+ \Big(\frac{(2\Delta x + C_1)^2}{2} + C_3\Big)f'(2\Delta x + x_i) - \Big(\frac{(\Delta x + C_1)^2}{2} + C_3\Big)f'(\Delta x + x_i)$$

$$\Big(\frac{(\Delta x - \frac{2}{3}\Delta x)^2}{2} + C_2\Big)f'(x_{i+1}) - \Big(\frac{(-\frac{2}{3}\Delta x)^2}{2} + C_2\Big)f'(x_i)$$
$$+ \Big(\frac{(2\Delta x - \frac{4}{3}\Delta x)^2}{2} + C_3\Big)f'(x_{i+2}) - \Big(\frac{(\Delta x - \frac{4}{3}\Delta x)^2}{2} + C_3\Big)f'(x_{i+1})$$

$$\Big(\frac{(\frac{\Delta x}{3})^2}{2}+C_2\Big)f'(x_{i+1})-\Big(\frac{(-\frac{2}{3}\Delta x)^2}{2}+C_2\Big)f'(x_i)$$
$$+\Big(\frac{(\frac{2\Delta x}{3})^2}{2}+C_3\Big)f'(x_{i+2})-\Big(\frac{(-\frac{\Delta x}{3})^2}{2}+C_3\Big)f'(x_{i+1})$$

$$\Big(\frac{\Delta x^2}{18}+C_2\Big)f'(x_{i+1})-\Big(\frac{4\Delta x^2}{18}+C_2\Big)f'(x_i)$$
$$+\Big(\frac{4\Delta x^2}{18}+C_3\Big)f'(x_{i+2})-\Big(\frac{\Delta x^2}{18}+C_3\Big)f'(x_{i+1})$$

$$\Big(C_2-C_3\Big)f'(x_{i+1})-\Big(\frac{4\Delta x^2}{18}+C_2\Big)f'(x_i)+\Big(\frac{4\Delta x^2}{18}+C_3\Big)f'(x_{i+2})$$

Let $C_3 = C_2$. This will make the $f(x_{i+1}$ term disappear, leaving the rest of the expression able to be factored into a coefficient in terms of C_2 and $(f(x_{i+2})-f(x_i))$. Setting this coefficient to equal zero and solving will give the value for both C_2 and C_3.

$$\Big(C_2-C_2\Big)f'(x_{i+1})-\Big(\frac{4\Delta x^2}{18}+C_2\Big)f'(x_i)+\Big(\frac{4\Delta x^2}{18}+C_2\Big)f'(x_{i+2})$$
$$\Big(\frac{2\Delta x^2}{9}+C_2\Big)f'(x_i)+\Big(\frac{2\Delta x^2}{9}+C_2\Big)f'(x_{i+2})$$
$$\Big(\frac{2\Delta x^2}{9}+C_2\Big)(f'(x_{i+2})-f(x_i))$$
$$\Big(\frac{2\Delta x^2}{9}+C_2\Big)=0$$
$$C_2=-\frac{2}{9}\Delta x^2$$

Because $C_3 = C_2$, we now know all four of the coefficients which allow for us to pull out N_n from an arbitrary function's integral.

$$C_0=-\frac{2}{3}\Delta x, C_1=-\frac{4}{3}\Delta x, C_2=-\frac{2}{9}\Delta x^2, C_3=-\frac{2}{9}\Delta x^2$$

Using these constants, we get the expression for the integral of $f(x)$ with a term related to N_n.

$$\int_{x_i}^{x_{i+2}} f(x)dx = N_n+\int_0^{\Delta x}\Big(\frac{(t+C_0)^2}{2}+C_2\Big)f''(t+x_i)dt$$
$$+\int_{\Delta x}^{2\Delta x}\Big(\frac{(t+C_1)^2}{2}+C_3\Big)f''(t+x_i)dt$$

$$\int_{x_i}^{x_{i+2}} f(x)dx = N_n + \int_0^{\Delta x} \Big(\frac{(t-\frac{2}{3}\Delta x)^2}{2} - \frac{2}{9}\Delta x^2\Big) f''(t+x_i)dt$$
$$+ \int_{\Delta x}^{2\Delta x} \Big(\frac{(t-\frac{4}{3}\Delta x)^2}{2} - \frac{2}{9}\Delta x^2\Big) f''(t+x_i)dt$$

$$\int_{x_i}^{x_{i+2}} f(x)dx = N_n + \int_0^{\Delta x} \Big(\frac{1}{2}t^2 - \frac{2}{3}\Delta x t\Big) f''(t+x_i)dt$$
$$+ \int_{\Delta x}^{2\Delta x} \Big(\frac{1}{2}t^2 - \frac{4}{3}\Delta x t + \frac{2}{3}\Delta x^2\Big) f''(t+x_i)dt$$

Note that the two integrals in terms of the second derivative of the original function $f''(x)$ are the error terms for our approximation over two sub-intervals. To evaluate the error for the entire interval, we need to sum the over every pair of sub-intervals. However, we will sum over every interval and divide the total by 2 because it has easier summation and still yields the same error bound. Additionally, we will take the absolute value of each term summed up so that total error is combined, even if $f(x) - \hat{f}(x)$ is negative.

$$|E_N| = \frac{1}{2}\sum_{i=0}^{n-1} \Big| \int_0^{\Delta x} \Big(\frac{1}{2}t^2 - \frac{2}{3}\Delta x t\Big) f''(t+x_i)dt$$
$$+ \int_{\Delta x}^{2\Delta x} \Big(\frac{1}{2}t^2 - \frac{4}{3}\Delta x t + \frac{2}{3}\Delta x^2\Big) f''(t+x_i)dt \Big|$$

$$|E_N| = \frac{1}{2}\sum_{i=0}^{n-1} \int_0^{\Delta x} \Big|\frac{1}{2}t^2 - \frac{2}{3}\Delta x t\Big| \Big|f''(t+x_i)\Big| dt$$
$$+ \int_{\Delta x}^{2\Delta x} \Big|\frac{1}{2}t^2 - \frac{4}{3}\Delta x t + \frac{2}{3}\Delta x^2\Big| \Big|f''(t+x_i)\Big| dt$$

Let K_2 be the smallest value such that $|f''(x)| \leq K_2$, $\forall x \in \mathbb{R}$. Importantly, because K_2 is not dependent on x, it is a constant. To use this important detail, we need to introduce it into our integral expressions.

First, we can change the order of summation and integration for both integrals because we know that the sum and integral must exist. Note that the only parts of our integral expressions that are dependent on the summation are the second derivative terms. Thus, the rest of the expressions can be treated as constants and can be factored out of the summation.

At this point, we can plug in our newly defined K_2, giving each term the expression $\sum_{i=0}^{n-1} K_2$. This can be simplified to nK_2 because K_2 is a constant in the summation and can then be factored out of the integrals.

$$|E_N| = \frac{1}{2}\Big(\int_0^{\Delta x} \Big|\frac{1}{2}t^2 - \frac{2}{3}\Delta x t\Big| \sum_{i=0}^{n-1} \Big|f''(t+x_i)\Big| dt$$
$$+ \int_{\Delta x}^{2\Delta x} \Big|\frac{1}{2}t^2 - \frac{4}{3}\Delta x t + \frac{2}{3}\Delta x^2\Big| \sum_{i=0}^{n-1} \Big|f''(t+x_i)\Big| dt\Big)$$

$$|E_N| \leq \frac{1}{2}\Big(\int_0^{\Delta x} \Big|\frac{1}{2}t^2 - \frac{2}{3}\Delta x t\Big| nK_2 dt + \int_{\Delta x}^{2\Delta x} \Big|\frac{1}{2}t^2 - \frac{4}{3}\Delta x t + \frac{2}{3}\Delta x^2\Big| nK_2 dt\Big)$$

$$|E_N| \leq \frac{nK_2}{2}\Big(\int_0^{\Delta x} \Big|\frac{1}{2}t^2 - \frac{2}{3}\Delta x t\Big| dt + \int_{\Delta x}^{2\Delta x} \Big|\frac{1}{2}t^2 - \frac{4}{3}\Delta x t + \frac{2}{3}\Delta x^2\Big| dt\Big)$$

To evaluate the integral of the absolute value, we need to see for which t values the expression inside is positive and negative. In our integral expressions, the insides of the absolute values are simple quadratic functions. We can find where these quadratics are positive and negative by simply factoring them.

First, we will factor the quadratic term for the integral from 0 to Δx. This will give us the values where the inside is zero from which we know that the values between the zeros will give a negative function value because the leading coefficient is positive.

$$\frac{1}{2}t^2 - \frac{2}{3}\Delta x t = 0$$
$$t(\frac{1}{2}t - \frac{2}{3}\Delta x) = 0$$
$$t = 0, \frac{1}{2}t - \frac{2}{3}\Delta x = 0$$
$$t = 0, \frac{4}{3}\Delta x$$

Note that the entire interval integrated over is within the factored t values. This means that the inside of the absolute value is negative for all integrated values. Therefore, we can rewrite the first integral to get rid of the absolute value.

$$\int_0^{\Delta x} \Big|\frac{1}{2}t^2 - \frac{2}{3}\Delta x t\Big| dt = -\int_0^{\Delta x} \Big(\frac{1}{2}t^2 - \frac{2}{3}\Delta x t\Big) dt$$

Similarly to the first integral, we will also solve for the zeros of the quadratic term of the second integral, giving us the interval for which the inside expression is positive in the same way we found it for the first integral.

$$\frac{1}{2}t^2 - \frac{4}{3}\Delta x t + \frac{2}{3}\Delta x^2 = 0$$
$$\therefore t = \frac{-(-\frac{4}{3}\Delta x) \pm \sqrt{(-\frac{4}{3}\Delta x)^2 - 4(\frac{1}{2})(\frac{2}{3}\Delta x^2)}}{2(\frac{1}{2})}$$

$$t = \frac{4}{3}\Delta x \pm \sqrt{\frac{16}{9}\Delta x^2 - \frac{4}{3}\Delta x^2}$$
$$t = \frac{4}{3}\Delta x \pm \sqrt{\frac{4}{9}\Delta x^2}$$
$$t = \frac{4}{3}\Delta x \pm \frac{2}{3}\Delta x$$
$$t = \frac{2}{3}\Delta x, 2\Delta x$$

Exactly like in the first integral expression, the entire interval integrated over is also within its factored t values. Therefore, we can rewrite the second integral to get rid of the absolute value in the same way we rewrote the first integral.

$$\int_{\Delta x}^{2\Delta x} \Big|\frac{1}{2}t^2 - \frac{4}{3}\Delta x t + \frac{2}{3}\Delta x^2\Big| dt = -\int_{\Delta x}^{2\Delta x} \Big(\frac{1}{2}t^2 - \frac{4}{3}\Delta x t + \frac{2}{3}\Delta x^2\Big) dt$$

Replacing the integrals with absolute value terms in the error bound formula with the equivalent expressions without absolute values gives a final integral expression that is easy to solve for an anti-derivative that can be evaluated.

$$|E_N| \leq \frac{nK_2}{2}\Bigg(-\int_0^{\Delta x} \Big(\frac{1}{2}t^2 - \frac{2}{3}\Delta x t\Big) dt - \int_{\Delta x}^{2\Delta x} \Big(\frac{1}{2}t^2 - \frac{4}{3}\Delta x t + \frac{2}{3}\Delta x^2\Big) dt \Bigg)$$
$$|E_N| \leq \frac{nK_2}{2}\Bigg(\int_0^{\Delta x} \Big(\frac{2}{3}\Delta x t - \frac{1}{2}t^2\Big) dt + \int_{\Delta x}^{2\Delta x} \Big(-\frac{1}{2}t^2 + \frac{4}{3}\Delta x t - \frac{2}{3}\Delta x^2\Big) dt \Bigg)$$
$$|E_N| \leq \frac{nK_2}{2}\Bigg(\Big(\frac{1}{3}\Delta x t^2 - \frac{1}{6}t^3\Big)\Big|_0^{\Delta x} + \Big(-\frac{1}{6}t^3 + \frac{2}{3}\Delta x t^2 - \frac{2}{3}\Delta x^2 t\Big)\Big|_{\Delta x}^{2\Delta x} \Bigg)$$

$$\begin{aligned} |E_N| \leq \frac{nK_2}{2}\Bigg(&\Big(\frac{1}{3}\Delta x(\Delta x)^2 - \frac{1}{6}\Delta x^3\Big) - \Big(\frac{1}{3}\Delta x(0)^2 - \frac{1}{6}(0)^3\Big) \\ &+ \Big(-\frac{1}{6}(2\Delta x)^3 + \frac{2}{3}\Delta x(2\Delta x)^2 - \frac{2}{3}\Delta x^2(2\Delta x)\Big) \\ &- \Big(-\frac{1}{6}\Delta x^3 + \frac{2}{3}\Delta x(\Delta x)^2 - \frac{2}{3}\Delta x^2(\Delta x)\Big)\Bigg) \end{aligned}$$

$$\begin{aligned} |E_N| \leq \frac{nK_2}{2}\Bigg(&\Big(\frac{1}{3}\Delta x^3 - \frac{1}{6}\Delta x^3\Big) + \Big(-\frac{4}{3}\Delta x^3 + \frac{8}{3}\Delta x^3 - \frac{4}{3}\Delta x^3\Big) \\ &- \Big(-\frac{1}{6}\Delta x^3 + \frac{2}{3}\Delta x^3 - \frac{2}{3}\Delta x^3\Big)\Bigg) \end{aligned}$$

$$|E_N| \leq \frac{nK_2}{2}\left(\frac{1}{6}\Delta x^3 + \frac{1}{6}\Delta x^3\right)$$

$$|E_N| \leq \frac{nK_2\Delta x^3}{6}$$

To remove the Δx^3 term, note that $\Delta x = \frac{x_n - x_0}{n}$. Thus, we can plug this expression into the error bound term.

$$|E_N| \leq \frac{K_2(x_n - x_0)^3}{6n^2}$$

Unfortunately, this error bound is somewhat large. When compared to the error bounds for other approximation techniques, this bound is larger than all of them. The following table shows the error bounds for this approximation, the Midpoint rule, the Trapezoidal rule, and Simpson's rule.

Approximation Technique	Error Bound
Our Approximation	$\frac{K_2(x_n - x_0)^3}{6n^2}$
Midpoint Rule	$\frac{K_2(x_n - x_0)^3}{24n^2}$
Trapezoidal Rule	$\frac{K_2(x_n - x_0)^3}{12n^2}$
Simpson's Rule	$\frac{K_4(x_n - x_0)^5}{180n^4}$

Fundamentally, the reason for this rather large bound on the error is because the linear regression is alike the Trapezoidal rule, only with half the number of partitions. While the least squares property is a useful one, it ultimately does not contribute to a better approximation, at least with this method. However, it should be noted that K_2 is not the same for each interval, but the maximum K_2 of all the intervals is applied to all of them. Practically, this means that it should be possible to find a better bound on the error, perhaps by using a different method.

4 Conclusion

The method outlined in this paper for deriving both the approximation formula and the error bound formula can be expanded to define a family of approximations, with one approximation for each order polynomial. Each of the higher order formulas operate on a similar logic to the linear regression formula. In each instance, the number of points used is always given by $n = d + 2$, where d denotes the order of the polynomial regression. Using Wolfram Mathematica, the formulas for numerical integration using higher-order polynomial regressions have been calculated using the method outlined for the linear case. The following table shows the recursive formulas for regression-based numerical integration using up to 8 points.

Number of Points	Approximation Expression (Recursive)
3	$\frac{2\Delta}{3}(f(x_i) + f(x_{i+1}) + f(x_{i+2}))$
4	$\frac{3\Delta}{8}(f(x_i) + 3f(x_{i+1}) + 3f(x_{i+2}) + f(x_{i+3}))$
5	$\frac{4\Delta}{105}(11f(x_i) + 26f(x_{i+1}) + 31f(x_{i+2}) + 26f(x_{i+3}) + 11f(x_{i+4}))$
6	$\frac{5\Delta}{288}(19f(x_i) + 75f(x_{i+1}) + 50f(x_{i+2}) + 50f(x_{i+3}) + 75f(x_{i+4}) + 19f(x_{i+5}))$
7	$\frac{\Delta}{770}(268f(x_i) + 933f(x_{i+1}) + 786f(x_{i+2}) + 646f(x_{i+3}) + 786f(x_{i+4}) + 933f(x_{i+5}) + 268f(x_{i+6}))$
8	$\frac{7\Delta}{17280}(751f(x_i) + 3577f(x_{i+1}) + 1323f(x_{i+2}) + 2989f(x_{i+3}) + 2989f(x_{i+4}) + 1323f(x_{i+5}) + 3577f(x_{i+6}) + 751f(x_{i+7}))$

Interestingly, using 4 points and finding the quadratic regression simply becomes Simpson's $\frac{3}{8}$ rule, though the other expressions do not show any discernible pattern. This highlights a counter-intuitive identity that the integral of the quadratic regression of 4 points is equal to the integral of the cubic interpolation of the points.

Additionally, one property of the approximation method which has not been used as of yet is the Least-Squares property, which is guaranteed for every formula in this family of approximations. Incorporating this property could give insight into a better error bound for the family of approximations.

One way in which this property could be incorporated into this finding is by changing our notion of distance. The notion of how far off our approximation is from the integral that we have used is the absolute value. If, however, one uses a different notion of distance, such as the L_2 norm, the least-squares property could be used to improve the error in some way.

References

[1] Edward A. Bender. Section 7.7 Deriving the Trapezoidal Rule Error. `https://mathweb.ucsd.edu/~ebender/20B/77_Trap.pdf`.

[2] Carnegie Mellon University. Error bounds on each approximation technique. `https://www.math.cmu.edu/~mittal/Recitation_notes.pdf`.